

CSI473 Phase 1 Design Report

UB Student Accommodation System — Team 14

Near-complete draft prepared for Laboratory 5 (interaction, lifecycle and integrated Phase 1 draft)

Assignment/phase	Phase 1
Approved project title	University of Botswana Student Accommodation System
Approval decision/date	Approved - 7 August 2026
Team number	14
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Repository URL	https://github.com/Neil-2003/CSI473_PROJECT.git
Submission tag	phase1-submission

1. Executive Summary

The University of Botswana Student Accommodation System addresses a genuine operational problem identified through stakeholder interviews and direct observation at the Student Accommodation Office. Currently accommodation applications are processed using manual methods involving paper forms and spreadsheets, causing delays for students and administrative burden for staff. Students have no clear way to track their application status, while administrators struggle to maintain accurate records of available rooms and current occupants.

This Phase 1 report presents the foundational design for a dedicated system that supports the application, allocation, and management of UB student accommodation. The system will enable students to submit accommodation applications online and view live status updates, while providing accommodation administrators with tools to manage rooms, process applications, and make allocations with 100% accuracy. The design excludes payment processing and external system integration to maintain a feasible semester-sized scope, as documented in Decision Record D-001.

The core vertical slice application submission → administrator review → room allocation → status notification has been modelled through requirements, use cases, domain analysis, quality scenarios, and interaction/lifecycle models. This draft integrates the evidence from Labs 2 through 5; architecture drivers and the remaining design chapters will be finalised for the Lab 6 / Phase 1 review.

2. Problem Evidence, Stakeholders and Scope

2.1 Problem Statement

The University of Botswana provides on-campus accommodation to students through university residences. Managing accommodation involves handling applications, tracking available spaces, allocating students to rooms and maintaining accurate occupancy records. Currently these processes are managed manually, leading to uncertainty for students and administrative burden for staff.

Students applying for accommodation often experience delays and confusion about their application status. They have no clear way to track whether their application has been received, approved or rejected. Meanwhile accommodation administrators struggle to maintain accurate records of available rooms, current occupants and pending applications across multiple residences.

When accommodation information is poorly managed, students may receive incorrect or late notifications about their accommodation status. Staff may inadvertently duplicate allocations, assign students to already-full rooms, or lose track of which students have been accommodated. This creates frustration for students and inefficiency for the university.

There is therefore a clear need for a dedicated system that supports the application, allocation, and management of UB student accommodation. Such a system would improve transparency, accuracy and efficiency for both students and accommodation staff.

2.2 Evidence

Evidence/source	Date/origin	What it shows
Student interviews (5 current UB students)	4–6 August 2026, UB Campus	Students reported waiting 2 to 3 weeks with no way to check their application status, causing repeated visits to the Accommodation Office.

Evidence/source	Date/origin	What it shows
Observation at the Student Accommodation Office	5 August 2026, Accommodation Office	Staff process applications using paper forms and spreadsheets; several students were queued asking staff for verbal status updates.
Review of the UB Accommodation Policy document	UB website, accessed 3 August 2026	Describes application deadlines and eligibility rules but makes no reference to an online application or tracking system.
Discussion with a Residence Hall Warden	6 August 2026, on-campus residence	Occupancy lists are maintained manually per residence; there is no centralised, real-time view of available rooms across residences.
Feedback from accommodation staff (2 staff members)	6 August 2026, Student Accommodation Office	Staff described difficulty cross-checking applications against room capacity and expressed interest in a digital system.

2.3 Approval Conditions

The project was approved on 7 August 2026 with the following conditions addressed:

- Payment processing excluded: the system boundary explicitly excludes payment processing and external system integration (see D-001). Students will continue to pay deposits through existing university channels.
- Student registration verification: administrators will manually verify student registration status using existing university data, as the system will not integrate with the university registry in Phase 1.
- Allocation rules: priority rules for allocation (first-year vs. continuing students, special requirements handling) will be defined and implemented in Phase 2 as part of the eligibility-checking workflow.

2.4 Goal and Objectives

Item	Team response
Goal	To develop a Student Accommodation System for the University of Botswana that reduces the time taken to process and communicate an accommodation application outcome from the current 2 to 3 weeks to a pilot target of under 72 hours, subject to administrative staff response time and availability.
Objective 1	Enable UB students to submit an accommodation application online and view a live application status (Submitted → Pending → Approved/Rejected) with a target status update within 48 hours of submission, dependent on timely administrator review.
Objective 2	Enable authorised accommodation staff to manage residence, room and occupancy records with 100% allocation accuracy, e.g. zero duplicate allocations and zero over-capacity allocations, verified through system validation rules.
Objective 3	Support allocation of eligible students to available rooms and automatically notify students of their application/allocation outcome,

Item	Team response
	targeting notification within 24 hours of a decision being recorded by an administrator.

2.5 Stakeholders

Stakeholder/role	Need	Concern	Evidence/access
UB Students	Apply for university accommodation and know their application/allocation status quickly and transparently.	Delays, unclear status, uncertainty about whether accommodation has been allocated.	5 student interviews, 4–6 Aug 2026.
UB Accommodation / Residence Staff	Manage applications, rooms, capacity and allocations efficiently.	Incorrect records, duplicate allocations, difficulty tracking available spaces.	Discussion with 2 staff members, 6 Aug 2026.
Accommodation Administrator	Approve/process applications and maintain accurate accommodation records.	Unauthorised changes and inaccurate accommodation information.	Direct observation of administrative functions, 5 Aug 2026.
Residence Hall Wardens	Visibility of occupancy in their residence and confirmation of student arrivals.	Inaccurate occupancy lists; students arriving without prior notification.	Discussion with a Residence Hall Warden, 6 Aug 2026.
System Administrator	Manage user accounts, permissions, and monitor system health.	Unauthorised access, data breaches, system downtime.	System administration and audit-trail requirements identified by the team.

2.6 Scope

In Scope

- UB student registration and login
- Student accommodation application
- Student profile and relevant accommodation information
- Viewing accommodation application status
- Management of UB residences
- Management of residence rooms
- Recording room capacity and occupancy
- Tracking available and occupied accommodation spaces
- Processing accommodation applications
- Allocating eligible students to available rooms

- Updating accommodation allocation status
- Administrator management of student accommodation records
- Role-based access for students and authorised accommodation staff
- Notifications / status messages for applications and allocations

Out of Scope

- Private / off-campus accommodation
- Accommodation provided by private landlords
- Property rental or property sales
- Rental payment processing
- Online banking / payment gateway integration
- Private accommodation advertisements
- Transport between residences and campus
- Property maintenance management unrelated to allocation
- Accommodation applications for universities other than UB
- External accommodation-provider management
- Real-time integration with the student registration system (unless separately approved)

2.7 Assumptions and Constraints

Assumptions	Constraints
The system is intended specifically for UB student accommodation.	Limited semester development time (12 weeks).
Students applying for accommodation are registered UB students.	Limited access to real UB accommodation data.
Accommodation staff have the authority to process applications and make allocations.	Limited access to accommodation administrators for requirements gathering.
Residence and room information is maintained by authorised university staff.	Real student records cannot be used unnecessarily during development.
The number of available spaces changes as students are allocated or leave accommodation.	The system will not integrate with real university systems unless specifically approved.
Synthetic / anonymised student information can be used during development and testing.	The project must remain within a feasible semester-sized scope.
The project will implement a manageable subset of the full UB accommodation process as its semester-sized vertical slice.	

3. Glossary

Term	Definition
Accommodation Application	A formal request submitted by a UB student for on-campus accommodation, containing student details and room preferences.

Term	Definition
Accommodation Administrator	Authorised university staff member responsible for reviewing applications, making allocations, and managing room/residence records.
Active Allocation	A current assignment of a student to a room. A student may have only one active allocation at any time.
Allocation	The assignment of an eligible student to a specific room, creating an association between the student and the room.
Application Status	The current state of an accommodation application: DRAFT, SUBMITTED, PENDING, APPROVED, or REJECTED.
Capacity	The maximum number of students that can be accommodated in a room.
Occupancy	The number of students currently allocated to a room. Must never exceed capacity.
Occupancy Record	A historical record tracking when a student occupied a specific room, including start and end dates.
Pending Application	An application that has been submitted and validated but not yet reviewed by an administrator.
Residence	A university-owned building containing rooms for student accommodation.
Room	An individual unit within a residence that can accommodate one or more students.
Student	A registered UB student who is eligible to apply for accommodation.
System Administrator	A user with permissions to manage user accounts and role-based permissions.
Vertical Slice	An end-to-end implementation of one core use case, from actor request through domain processing to response.

4. Functional Requirements and Use Cases

4.1 Functional Requirements

ID	Functional Requirement
FR-01	The system shall allow a registered UB student to log in using their student account credentials.
FR-02	The system shall allow a logged-in UB student to submit an accommodation application, capturing the student's details and accommodation preferences.
FR-03	The system shall validate a submitted accommodation application and record it with a Pending status.
FR-04	The system shall allow a student to view the current status of their application.
FR-05 (Revised)	The system shall allow an authorised accommodation administrator to view a paginated list of submitted accommodation applications. The list must be

ID	Functional Requirement
	sortable by submission date and filterable by status (e.g., Pending, Approved, Rejected). If no applications match the filter criteria, the system shall display an informative message indicating that the list is empty.
FR-06	The system shall allow an authorised accommodation administrator to review a Pending application against room availability and eligibility rules.
FR-07	The system shall allow an authorised accommodation administrator to allocate an eligible student to an available room.
FR-08	The system shall reject an allocation attempt and notify the administrator when the selected room has no remaining capacity.
FR-09	The system shall prevent a student from holding more than one active accommodation allocation at the same time.
FR-10	The system shall update a room's occupancy record whenever a student is allocated to, or removed from, that room.
FR-11	The system shall allow an authorised accommodation administrator to record an application as Rejected/Unsuccessful and also state the reason.
FR-12	The system shall allow an authorised accommodation administrator to add, update, or deactivate residence and room records including room capacity.
FR-13	The system shall allow an authorised system administrator to manage user accounts and assign role-based permissions.

4.2 Actors and Goals

Actor	Type	Goal
UB Student	Primary	Apply for university accommodation and know their application/allocation status.
Accommodation Administrator	Primary	Process accommodation applications and manage room allocations and residence records.
System Administrator	Primary	Manage system user accounts and role-based permissions.
Residence Hall Wardens	Primary	See accurate, current occupancy for their residence.
Residence Staff	Primary	Manage applications, rooms, capacity and allocation efficiently.

4.3 Use Cases

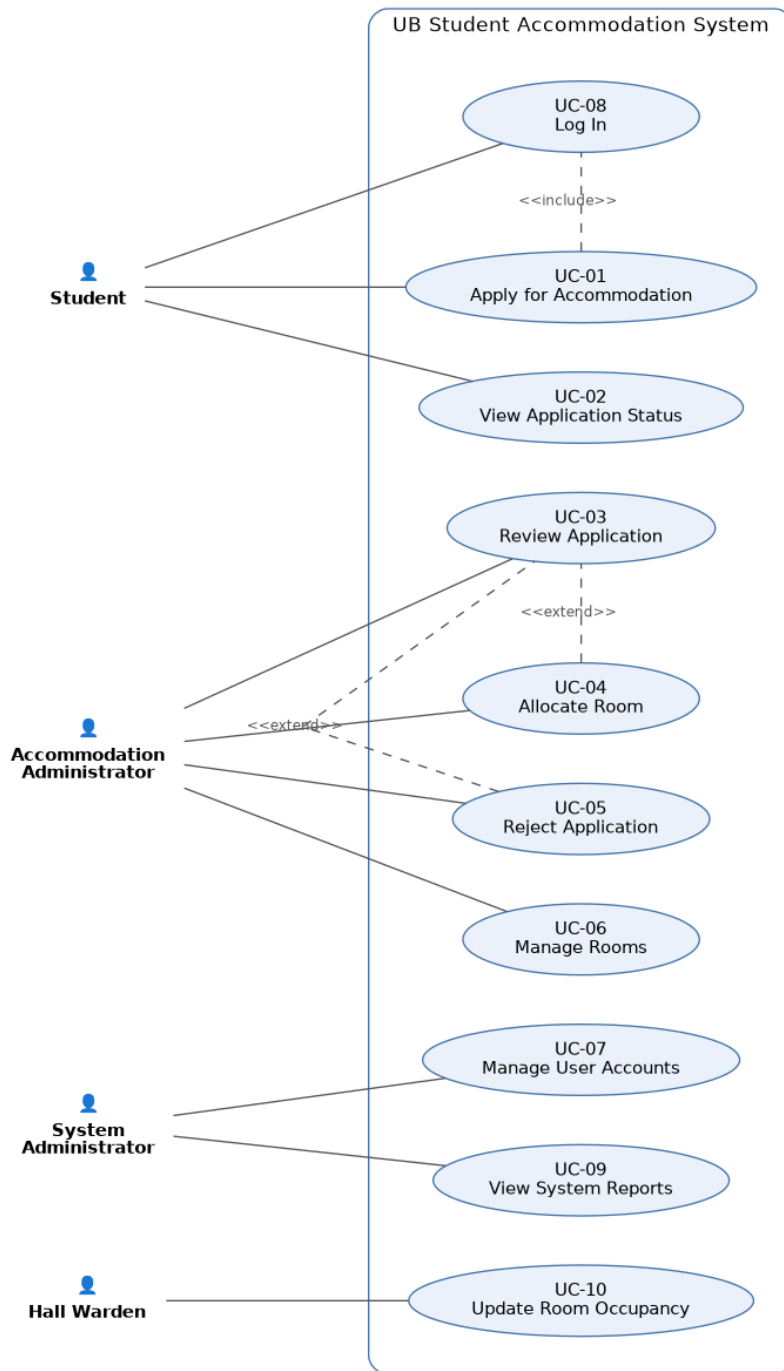
ID	Use Case	Primary Actor
UC-01	Apply for Accommodation	Student
UC-02	View Application Status	Student
UC-03	Review Application	Accommodation Administrator
UC-04	Allocate Room	Accommodation Administrator

ID	Use Case	Primary Actor
UC-05	Reject Application	Accommodation Administrator
UC-06	Manage Rooms	Accommodation Administrator
UC-07	Manage User Accounts	System Administrator
UC-08	Log In	Student
UC-09	View System Reports	System Administrator
UC-10	Update Room Occupancy	Hall Warden

System boundary: UB Student Accommodation System. UC-01 is the fully dressed core use case, covering the vertical slice: submission → review → allocation/rejection → status update.

4.4 Use-Case Diagram

The use-case diagram below illustrates the system boundary, actors, and their interactions with the UB Student Accommodation System.



Use-case diagram (editable source: models/use-case-diagram.dot)

5. Quality Requirements

The following measurable quality scenarios were derived from the stakeholder evidence in Section 2 and the objectives in Section 2.4. Each scenario follows the source–stimulus–artefact–environment–response–measure pattern and is traced to at least one functional requirement or business rule.

ID	Quality attribute	Scenario (source / stimulus / artefact / environment / response / measure)
QS-01	Performance	Source: Student. Stimulus: requests their application status list. Artefact: status view. Environment: normal operation, up to 100

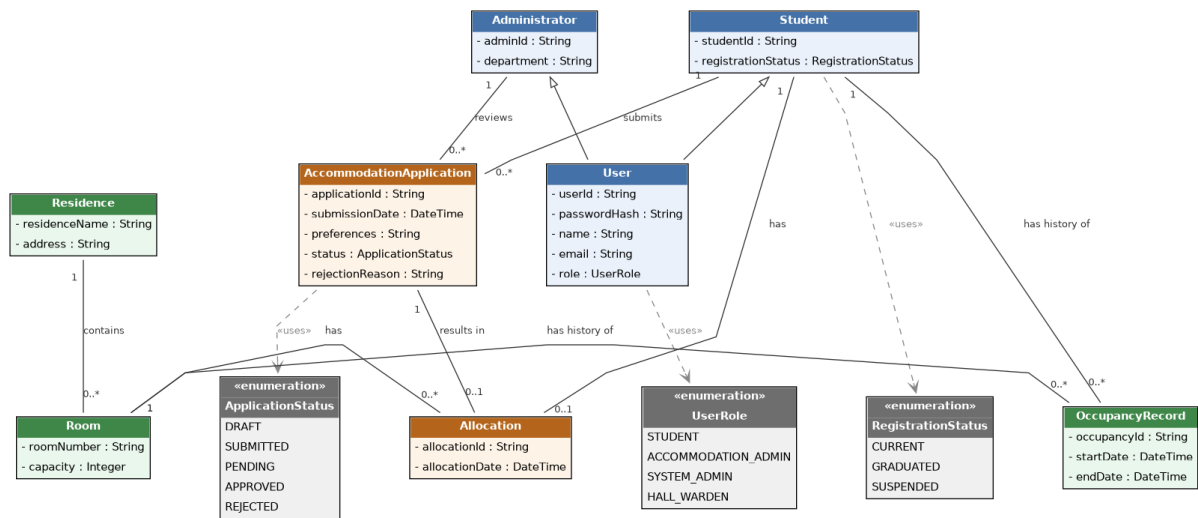
ID	Quality attribute	Scenario (source / stimulus / artefact / environment / response / measure)
		concurrent users during peak registration week. Response: status is retrieved and displayed. Measure: within 2 seconds for 95% of requests. Related: FR-04.
QS-02	Availability	Source: Any authenticated user. Stimulus: accesses the system during the weekday application window (08:00–17:00). Artefact: whole system. Environment: normal operation. Response: system responds to requests. Measure: at least 99% uptime during the stated window over the semester pilot. Related: Objective 1.
QS-03	Data integrity / reliability	Source: Accommodation Administrator. Stimulus: attempts to allocate a student to a room that is already at capacity. Artefact: Allocation / Room. Environment: normal operation. Response: allocation is rejected and the administrator is notified. Measure: 100% of over-capacity allocation attempts are rejected (zero over-capacity allocations in testing). Related: FR-08, BR-02.
QS-04	Security / authorisation	Source: An unauthenticated or student-role user. Stimulus: attempts to access an administrator-only function (e.g. allocate a room). Artefact: role-based access control. Environment: normal operation. Response: access is denied and the attempt is logged. Measure: 100% of unauthorised attempts are blocked, verified across all administrator functions. Related: FR-13, BR-04.
QS-05	Usability	Source: A newly authorised accommodation administrator. Stimulus: is asked to locate and filter the list of Pending applications. Artefact: administrator application list (FR-05). Environment: first use, no prior training beyond a one-page quick guide. Response: administrator locates and filters the list unaided. Measure: task completed within 3 minutes by at least 4 of 5 test users. Related: FR-05 (Revised).

These scenarios will inform the architecture drivers to be finalised in Lab 6/7 (Section 8); QS-03 and QS-04 in particular constrain the domain and interaction design already modelled in Sections 6 and 7.

6. Analysis/Domain Model and Responsibilities

This section summarises the domain analysis produced in Laboratory 4. Full evidence is committed at `models/domain-model.pdf` (+ `.dot` source), `docs/business-rules.md`, `docs/crc-cards.md`, `docs/traceability-matrix.md` and `decisions/D-001.md`, `D-002.md`.

6.1 Domain Class Model



Domain class model (editable source: `models/domain-model.dot`)

The model represents the problem-domain concepts only (no screens, controllers or database tables): User specialises into Student and Administrator; a Student submits an AccommodationApplication that an Administrator reviews; an approved application results in an Allocation of a Student to a Room within a Residence; each Room accumulates a history of OccupancyRecord entries. Enumerations (UserRole, ApplicationStatus, RegistrationStatus) constrain the valid values of key attributes.

6.2 Business Rules and Invariants

ID	Rule (summary)	Source
BR-01	A student must not hold more than one active accommodation allocation at the same time.	FR-09, UC-01 (A3)
BR-02	The number of students allocated to a room must never exceed the room's capacity.	FR-08, UC-01 (A2)
BR-03	An application must follow the life cycle DRAFT → SUBMITTED → PENDING → APPROVED/REJECTED.	FR-03, FR-11
BR-04	Only ACCOMMODATION_ADMIN or SYSTEM_ADMIN roles may review, allocate, reject or manage rooms.	FR-05, FR-06, FR-07, FR-12
BR-05	Only registered UB students may submit an accommodation application.	FR-01, FR-02
BR-06	A room's occupancy must always be consistent with its active allocations.	FR-10

6.3 Responsibility Allocation (CRC Summary)

Class	Key responsibilities	Key collaborators
AccommodationApplication	Create and store itself; track/manage status; validate completeness; record rejection reason; request an allocation.	Student, Administrator, Allocation

Class	Key responsibilities	Key collaborators
Room	Store its details; calculate occupancy from active allocations; refuse an allocation when full.	Residence, Allocation
Allocation	Create itself for a student/room pair; enforce BR-01 and BR-02; record the allocation date; end an allocation.	Student, Room, OccupancyRecord
Administrator	Review pending applications; make allocation decisions; reject applications; manage room/residence records.	AccommodationApplication, Allocation, Room
OccupancyRecord	Store occupancy history; open/close a record as allocations start/end.	Room, Allocation

6.4 Design Rationale

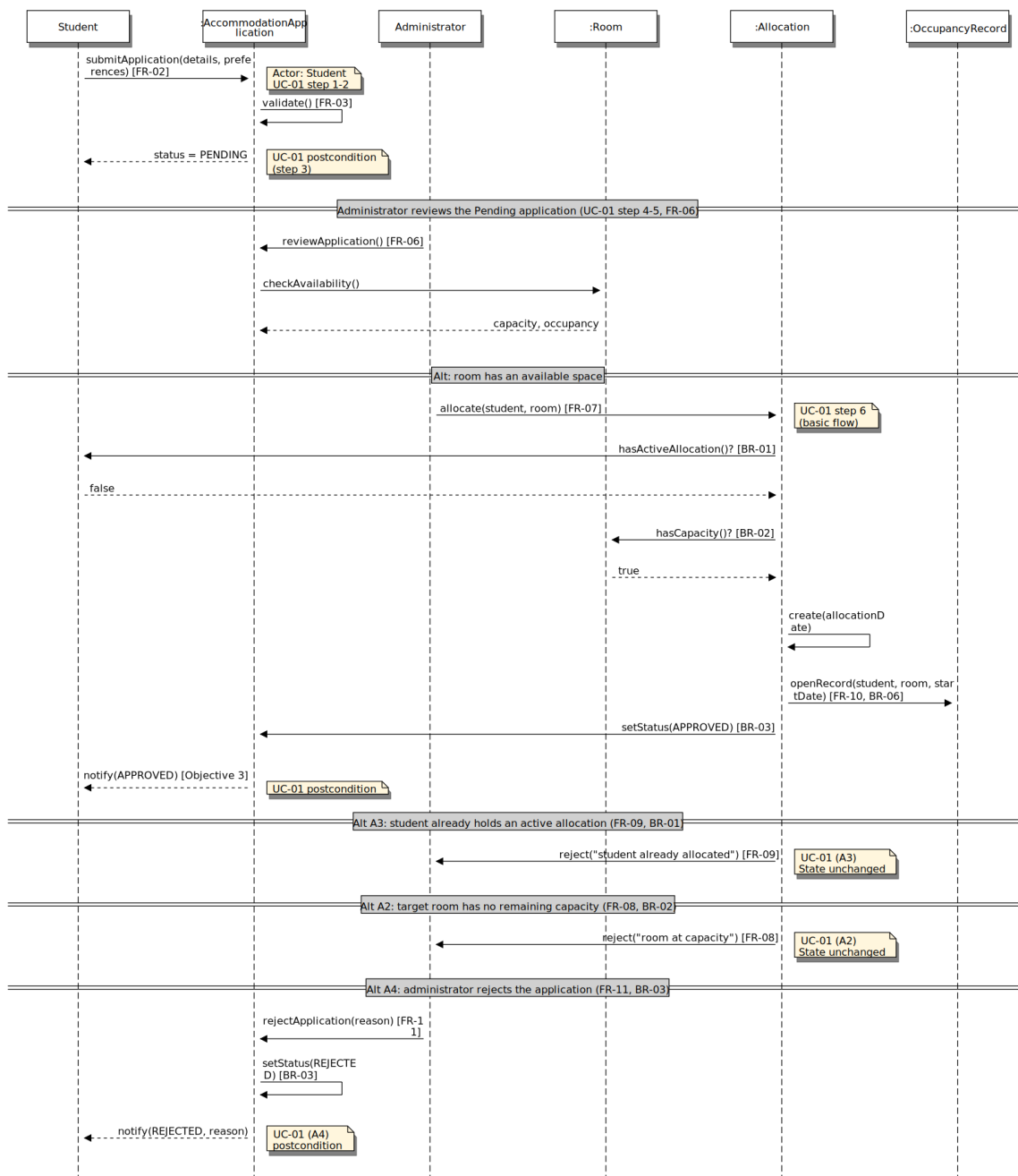
Two key modelling decisions were recorded (decisions/D-001.md, D-002.md):

- D-001 — Occupancy is derived from active Allocation records rather than stored as a counter on Room, avoiding a cache-style value that could drift out of sync with BR-06.
- D-002 — A separate Allocation class represents the student–room relationship rather than a roomId attribute on Student, enabling accommodation history, audit information, and a clean enforcement point for BR-01.

7. Interaction and Lifecycle

This section presents the Laboratory 5 evidence: a sequence diagram for the core use case (UC-01, including its basic flow and alternatives A2–A4), and the lifecycle model for the state-sensitive AccommodationApplication entity. Full evidence is committed at models/sequence-core-use-case.*, models/lifecycle-or-activity.* and docs/consistency-matrix.md.

7.1 Sequence Diagram — UC-01 Apply for Accommodation



Sequence diagram for UC-01, basic flow plus alternatives A2 (room full), A3 (duplicate allocation) and A4 (rejection).

The diagram traces the full vertical slice: a Student submits an application to the AccommodationApplication object (FR-02, FR-03); an Administrator reviews it and checks Room availability (FR-06); on the basic flow the Administrator allocates the student, and the Allocation object itself enforces BR-01 (no duplicate active allocation) and BR-02 (room capacity) before creating the allocation, opening an OccupancyRecord (FR-10, BR-06) and updating the application's status to APPROVED (BR-03). The three alternative flows show the two rule-violation rejections (A2, A3) required by FR-08/FR-09, and the administrator-initiated rejection path (A4) required by FR-11.

7.2 Lifecycle — AccommodationApplication

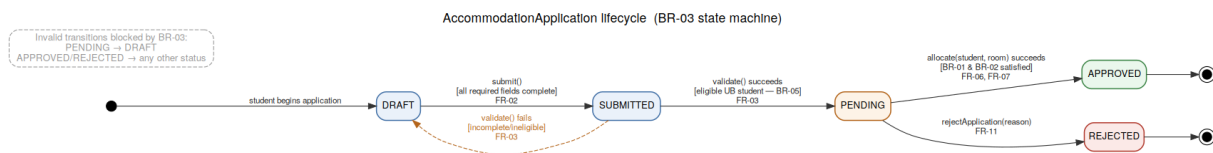


Figure 7.2 — AccommodationApplication state machine (editable source: `models/lifecycle-or-activity.dot`)

The state machine implements BR-03 directly: an application starts in DRAFT, moves to SUBMITTED once the student submits it (FR-02), and to PENDING once validation succeeds (FR-03); failed validation returns it to DRAFT. From PENDING, a successful allocation moves it to APPROVED (FR-07) and an administrator rejection moves it to REJECTED (FR-11); both are terminal states. The model explicitly blocks the invalid transitions named in BR-03 (PENDING → DRAFT directly, and any transition out of APPROVED or REJECTED).

7.3 Consistency Check

A full step-by-step consistency matrix (`docs/consistency-matrix.md`) links every UC-01 step to its sequence-diagram message, the responsible CRC class, the resulting application state, and the requirement/business rule it satisfies. No new classes, states or actors were introduced during interaction modelling that are not already present in the domain model (Section 6), the use-case table (Section 4.3) or the CRC cards — the one contradiction found and corrected during this pass is recorded in `evidence/lab-05-review.md`.

8. Architecture and Decisions

[To be completed for Lab 6/7 — 25 September 2026. Architecture drivers will be derived primarily from QS-01 to QS-05 (Section 5) and BR-01/BR-02/BR-04 (Section 6.2). A first candidate driver: the allocation workflow must enforce BR-01 and BR-02 atomically to satisfy QS-03, which will shape the choice between a single allocation service and a distributed transaction approach.]

9. Data, API/Service and Deployment Design

[To be completed for Lab 8 — 2 October 2026, per the project milestone schedule.]

10. Detailed Design, Patterns and Framework

[To be completed for Lab 9 — 9 October 2026, per the project milestone schedule.]

11. User Interface, Accessibility and Usability

[To be completed for Lab 10 — 23 October 2026. Will incorporate QS-05 usability evidence from Section 5.]

12. Implementation Vertical Slice

[To be completed for Lab 11 / Phase 2 — 30 October 2026. The approved core use case for the vertical slice is UC-01 (Apply for Accommodation), covering submission through allocation/rejection, as modelled in Section 7.]